

7824 Zero Road, second floor

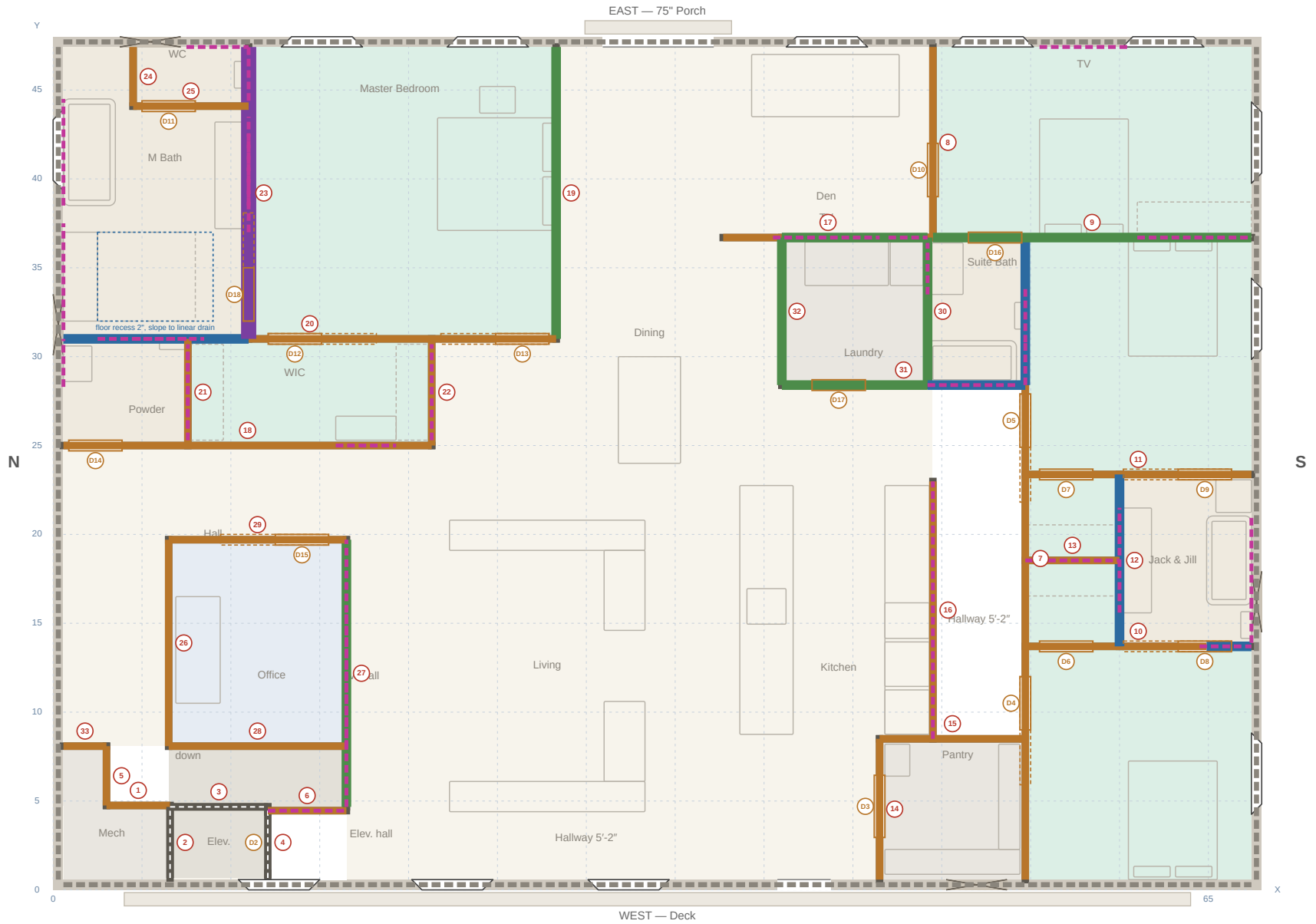
S-1 Framing plan · packet for the framing contractor

Interior partitions on the O3B second floor: wall types, door rough openings, headers, blocking, and the exterior furring as given.

Rev C · 2026-10-01 · Schematic for pricing and coordination, not a stamped construction document. Verify every dimension on site. Coordinates are feet from the north and west brick faces; the plan is drawn with east at the top and north on the left. Wall numbers match the concept drawing. Questions: name the sheet and the tag (for example S-1 W23).

Contents

1. S-1 sheet
2. Notes and open items
3. Schedules
4. Bill of materials



7824 Zero Road — second floor

Casper, Wyoming · 68' x 48' to brick, faces 7" inside · 9' ceiling · grid 5'
Wall numbers as the floor plan drawing of 2026-09-29

Framing plan

Interior partitions: wall types, door rough openings, headers and blocking

Schematic for pricing and coordination, not a stamped construction document. Verify on site.

S-1

rev C · 2026-10-01

drawn for K. Christianson

Notes

1. Positions are taped from the nearest wall framing, exterior walls first. Interior walls: to the face of the studs; W# is the wall number on S-1. Exterior walls: to the face of the exterior studs as framed, 6" in from the brick. "On W12" means the item sits in that wall, and the distance runs along it. Wet walls are dimensioned to their 2x4 line: on the side a 2x6 or 2x10 grows toward, its face sits 1-1/2" (2x10: 5-1/4") closer.
2. All interior walls are non-bearing partitions under a clear-span 9'-0" flat ceiling. Headers: a single flat 2x4 for every opening up to 8' (IRC R602.7.4), cripples at 16" to the double top plate. Pocket doors use the frame maker's header.
3. Ceiling: roof trusses run east-west (along Y) at 24" o.c. Partitions that run east-west sit between trusses and need 2x6 flat blocking at 24" between the bottom chords to catch the top plate; partitions that run north-south cross a truss every 24". Hang every top plate with slip clips (1" gap, no nails through the plate into the chord) to avoid truss uplift cracking.
4. Walls are drawn at 5" (2x4 + 1/2" board each side). A 2x6 wet wall is 6-1/2"; the extra 1-1/2" grows toward the side named in the schedule, so the room on the other side keeps its drawn dimension.
5. Doors: 8'-0" leaves. Swing door RO = leaf + 2" wide × 98-1/2" tall. Pocket door RO = 2 × leaf + 1" wide × 100-1/2" tall; the dashed box shows the pocket side. D3 is a 42" cased opening. D4, D5 (36" pockets) and D8, D9 each have a partition abutting the pocket side of the wall (walls 15, 11 and 12): the abutting wall nails to the pocket frame's reinforced split studs, not through the cavity.
6. Staggered-stud walls (green): 2x6 plates, 2x4 studs alternating faces at 12" o.c. (24" o.c. on each face), R-19 batt woven between, no stud touches both faces. Pipes and the washer box thread between the studs; the dryer gets a recessed dryer box in wall 17. 2x10 wall 23 (purple): 2x10 plates and studs at 16" o.c.; the pocket frame for D18 sits in the bedroom half, the vanity drains and the WC vent in the bath half.
7. Exterior furring (grey dashed): 2x4 @ 24" o.c. in the 7" band, standing 1/2" off the brick as an air gap. Shown in the 3D model; nothing changes there except the blocking listed.
8. Every door has at least 32" clear and 18" latch-side clearance on the pull side per the plan; keep thresholds at 1/2" max (1/4" at the shower, which has none).
9. Curbless shower (2'-0" to 7'-6" from the north ext. wall; across, 10'-6" to 15'-6" from east ext. wall): recess the floor 2" over the pad and 12" beyond its open side, slope 1/4" per foot to the linear drain on the bench side.
10. Blocking is shown as magenta bands on the face noted. Grab-bar blocking is continuous 2x8 flat with its centre at 34" AFF; vanity blocking 2x10 flat 26"–34" AFF (wall-hung roll-under vanities, 27" clear beneath); TV blocking 3/4" plywood 42"–66" AFF.
11. Sound: R-13 batts in every bedroom and bathroom wall and in walls 7, 9, 17, 19, 20 and 23.
12. Exterior walls: brick shell with the measured 7" furring. New blocking in the furring is shown on the sheet; nothing structural changes there. No water piping in the furring (see P-1).
13. Elevator hoistway walls (2, 3, 4) and the stair opening are as measured; the stair guard and handrails follow IRC R311.7.8 and R312 (34"–38" rail, 36" guard, 4" sphere).
14. Sub-panel P2 sits on the landing face of wall 1 so it stays reachable.

Open items

- Pocket doors D4, D5, D8, D9: each pocket crosses a T-junction (walls 15, 11, 12). Workable with the frame's steel split studs; confirm, or swap D4 and D5 back to swings.
- Door leaf style and hardware: lever handles, 42" leaves for Bedrooms 3 and 4, and whether the elevator-hall side of wall 6 is a full wall or a guard.
- Confirm the wet-wall growth direction on walls 7, 17, 20 and 31 (each steals 1-1/2" from the bath or laundry side).
- 2nd Suite wardrobe: a 2' deep reach-in needs a front wall and doors that are not on the plan; confirm, or keep it as freestanding wardrobes.

Schedules

Wall schedule

Wall	Runs	Length	Framing	2x6 grows	Doors	Notes
W1	north–south, W5 to W2; west face 4'-1" from west ext. wall	3'-7"	2x4 partition			Mech room, landing side (W5 to the hoistway). 3/4" plywood backing 48" × 48" on the landing face for sub-panel P2. The mech room and the vent leg are one L-shaped room with no door.
W2	east–west, west ext. wall to W1; north face 5'-11" from north ext. wall	4'-0"	existing hoistway			Hoistway, west.
W3	north–south, W2 to W4; west face 4'-0" from west ext. wall	5'-6"	existing hoistway			Hoistway, south, with the 3'-0" car door opening to the elevator hall.
W4	east–west, west ext. wall to W3; north face 5'-2" from W2	4'-0"	existing hoistway		D2 Elevator car door	Hoistway, east.
W5	east–west, W1 to W33; north face 2'-4" from north ext. wall	3'-4"	2x4 partition			Mech closet leg (vents).
W6	north–south, W4 to W27; west face 3'-10" from west ext. wall	4'-5"	2x4 partition			Elevator hall to stair. Full height; stair side gets the handrail blocking.
W7	east–west, west ext. wall to W9; south face 12'-8" from south ext. wall	36'-1"	from the west ext. wall end: 2x4 partition 27'-10", then 2x6 wet wall 8'-4"	west	D4 Bedroom 3, D5 Bedroom 4	Wing wall, 36'-1". 2x6 for the last 8'-4" at the W9 end, for the suite-bath toilet vent; grows into the bath.
W8	east–west, W9 to east ext. wall; south face 17'-10" from south ext. wall	10'-9"	2x4 partition		D10 2nd Suite	2nd Suite west wall.
W9	north–south, W8 to south ext. wall; east face 10'-8" from east ext. wall	17'-11"	2x6 plate, staggered 2x4 studs	north	D16 Suite bath	2nd Suite / Bedroom 4. Staggered studs for sound; headboards both sides. Grows into Bedroom 4.

Wall	Runs	Length	Framing	2x6 grows	Doors	Notes
W10	north–south, W7 to south ext. wall; west face 13'-0" from west ext. wall	12'-9"	from the W7 end: 2x4 partition 10'-2", then 2x6 wet wall 2'-6"	south	D6 Closet 3, D8 Jack & Jill	Bedroom 3 / closet 3 / Jack & Jill. 2x6 for the last 2'-6" at the south ext. wall, for the tub valve; grows into the bath.
W11	north–south, W7 to south ext. wall; west face 4'-6" from W13	12'-9"	2x4 partition		D7 Closet 4, D9 Jack & Jill	Bedroom 4 / closet 4 / Jack & Jill.
W12	east–west, W10 to W11; south face 7'-4" from south ext. wall	9'-8"	2x6 wet wall	west		Jack & Jill vanity wall: two lav drains and stack S4. Grows into closets 3 and 4.
W13	north–south, W7 to W12; east face 4'-6" from W11	5'-4"	2x4 partition			Between closets 3 and 4.
W14	east–west, west ext. wall to W15; south face 7'-10" from W7	7'-11"	2x4 partition		D3 Pantry	Pantry west wall, hallway side.
W15	north–south, W14 to W7; west face 7'-10" from west ext. wall	8'-2"	2x4 partition			Pantry / kitchen.
W16	east–west, W15 to a free end 5'-3" short of W31; north face 48'-10" from north ext. wall	14'-6"	2x4 partition			Kitchen wall run. Continuous 3/4" plywood backing 0–96" AFF on the kitchen face for tall units, oven tower and hood; stack S2 (2") inside.
W17	north–south, a free end 9'-3" short of W19 to W8; east face 10'-8" from east ext. wall	11'-10"	from the free end: 2x4 partition 3'-4", then 2x6 plate, staggered 2x4 studs 8'-2", then 2x4 partition 4"	north		Den / laundry. Staggered-stud sound wall along the laundry (washer box and laundry sink thread between the studs; recessed dryer box); grows into the laundry. TV backing on the den face, 2'-10"–8'-10" from W8.
W18	north–south, north ext. wall to W22; west face 24'-4" from west ext. wall	20'-9"	2x4 partition		D14 Powder	Hall / powder and walk-in.
W19	east–west, W20 to east ext. wall; south face 12'-4" from W32	16'-5"	2x6 plate, staggered 2x4 studs	west		Master bedroom east wall; headboard wall. Staggered studs for sound; grows into the bedroom.
W20	north–south, north ext. wall to W19; east face 16'-4" from east ext. wall	27'-9"	from the north ext. wall end: 2x6 wet wall 10'-5", then 2x4 partition 17'-4"	south	D12 WIC, D13 Master bedroom	Master bath and bedroom south wall. 2x6 from the north ext. wall to W23 for the powder toilet, shower valve and stack S1; grows into the powder room and walk-in.
W21	east–west, W18 to W20; north face 6'-11" from north ext. wall	6'-0"	2x4 partition			Powder / walk-in.
W22	east–west, W18 to W20; north face 13'-5" from W21	6'-0"	2x4 partition			Walk-in east wall. Rod and shelf backing.
W23	east–west, W20 to east ext. wall; north face 10'-4" from north ext. wall	16'-5"	2x10 wall	east	D18 Master bath	Bath / bedroom wall, 2x10: pocket door D18 at the south end, the double vanity drains, the WC toilet vent and the TV wiring share it. Grows into the bedroom. TV backing on the bedroom face.

Wall	Runs	Length	Framing	2x6 grows	Doors	Notes
W24	east–west, W25 to east ext. wall; north face 3'-10" from north ext. wall	3'-4"	2x4 partition			WC west wall. The tub beyond is freestanding; no valve in this wall.
W25	north–south, W24 to W23; east face 3'-3" from east ext. wall	6'-6"	2x4 partition		D11 WC	WC door wall.
W26	east–west, W28 to W29; north face 5'-10" from north ext. wall	11'-7"	2x4 partition			Office west wall.
W27	east–west, W6 to W29; north face 9'-8" from W26	15'-0"	2x6 plate, staggered 2x4 studs	west		Office and stair east wall. Staggered studs for sound; grows into the office and stair. TV backing on the living face, 7'-6"–15'-6" from west ext. wall; handrail blocking on the stair face.
W28	north–south, W26 to W27; west face 3'-1" from W3	10'-0"	2x4 partition			Stair / office.
W29	north–south, W26 to W27; east face 5'-0" from W18	10'-0"	2x4 partition		D15 Office	Office north wall with the office door.
W30	east–west, W31 to W17; south face 5'-2" from W7	8'-4"	2x6 plate, staggered 2x4 studs	west		Laundry / suite bath: staggered-stud sound wall; the suite vanity drain and vent stack S3 thread between the studs. Grows into the laundry.
W31	north–south, W32 to W7; west face 27'-9" from west ext. wall	13'-8"	from the W32 end: 2x6 plate, staggered 2x4 studs 8'-2", then 2x6 wet wall 5'-6"	north	D17 Laundry	Hall / laundry and suite bath. Staggered studs along the laundry, 2x6 behind the suite tub for its valve; grows into the laundry and bath.
W32	east–west, W31 to W17; south face 7'-10" from W30	8'-4"	2x6 plate, staggered 2x4 studs	east		Laundry west wall, dining side, moved north 1'. Staggered studs for sound; grows into the laundry.
W33	north–south, north ext. wall to W5; west face 7'-5" from west ext. wall	2'-5"	2x4 partition			Vent leg, passage side: a wall where door D1 was.

Door schedule

Door	Room	From	Wall	Leaf	Type	Rough opening	Header
D2	Elevator car door	elevator hall	W4, RO 8"–3'-8" from west ext. wall	3'-0" car door	hoistway	by elevator contractor	—
D3	Pantry	hallway	W14, RO 2'-4"–6'-0" from west ext. wall	3'-6" cased opening, 8'-0"	opening	44" × 98.5"	flat 2x4
D4	Bedroom 3	wing hallway	W7, RO 5'-5"–11'-6" from west ext. wall	3'-0" × 8'-0"	pocket to the west	73" × 100.5"	frame kit
D5	Bedroom 4	wing hallway	W7, RO 4"–6'-5" from W31	3'-0" × 8'-0"	pocket to the west	73" × 100.5"	frame kit
D6	Closet 3	Bedroom 3	W10, RO 8'-11"–12'-1" from south ext. wall	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D7	Closet 4	Bedroom 4	W11, RO 8'-11"–12'-1" from south ext. wall	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D8	Jack & Jill	Bedroom 3	W10, RO 1'-2"–7'-3" from south ext. wall	3'-0" × 8'-0"	pocket to the north	73" × 100.5"	frame kit

Door	Room	From	Wall	Leaf	Type	Rough opening	Header
D9	Jack & Jill	Bedroom 4	W11, RO 1'-2"–7'-3" from south ext. wall	3'-0" × 8'-0"	pocket to the north	73" × 100.5"	frame kit
D10	2nd Suite	den	W8, RO 5'-5"–8'-7" from east ext. wall	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D11	WC	master bath	W25, RO 4'-5"–7'-7" from north ext. wall	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D12	WIC	master bedroom	W20, RO 11"–7'-0" from W23	3'-0" × 8'-0"	pocket to the south	73" × 100.5"	frame kit
D13	Master bedroom	open corner	W20, RO 3"–6'-4" from W19	3'-0" × 8'-0"	pocket to the north	73" × 100.5"	frame kit
D14	Powder	hall	W18, RO 4"–3'-6" from north ext. wall	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D15	Office	hall	W29, RO 8'-11"–15'-0" from north ext. wall	3'-0" × 8'-0"	pocket to the north	73" × 100.5"	frame kit
D16	Suite bath	2nd Suite	W9, RO 12'-11"–16'-1" from south ext. wall	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D17	Laundry	dining	W31, RO 1'-5"–4'-7" from W32	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D18	Master bath	master bedroom	W23, RO 9'-5"–15'-6" from east ext. wall	3'-0" × 8'-0"	pocket to the east	73" × 100.5"	frame kit

Blocking schedule

Tag	Wall	Span	Blocking	For
B1	W20	2'-0"–8'-0" from north ext. wall	grab bar: 2x8 flat, centre 34" AFF	master shower, back wall (bath face)
B2	W20	4'-0"–7'-6" from north ext. wall	grab bar: 2x8 flat, centre 34" AFF	powder toilet, rear (powder face)
B3	W21	4"–4'-10" from W20	grab bar: 2x8 flat, centre 34" AFF	powder toilet, side (powder face)
B4	W23	1"–3'-1" from east ext. wall	grab bar: 2x8 flat, centre 34" AFF	WC toilet, rear (WC face)
B5	W23	4'-0"–10'-6" from east ext. wall	wall-hung vanity: 2x10 flat, 26"–34" AFF	master double vanity, roll-under (bath face)
B6	W23	5'-0"–9'-0" from east ext. wall	TV: 3/4" ply 24" tall, 42"–66" AFF	master bedroom TV (bedroom face)
B7	W7	2'-3"–5'-3" from W31	grab bar: 2x8 flat, centre 34" AFF	suite toilet, rear (bath face)
B8	W31	2"–4'-10" from W30	grab bar: 2x8 flat, centre 34" AFF	suite tub, long side (bath face)
B9	W7	2"–2'-4" from W31	grab bar: 2x8 flat, centre 34" AFF	suite tub, end (bath face)
B10	W30	2"–3'-0" from W17	wall-hung vanity: 2x10 flat, 26"–34" AFF	suite vanity, roll-under (bath face)
B11	W17	2"–2'-0" from W30	wall-hung vanity: 2x10 flat, 26"–34" AFF	laundry sink (laundry face)
B12	W17	2'-6"–8'-6" from W30	TV: 3/4" ply 24" tall, 42"–66" AFF	den TV (den face)
B13	W27	7'-6"–15'-6" from west ext. wall	TV: 3/4" ply 24" tall, 42"–66" AFF	living TV (living face)
B14	W12	1'-8"–7'-7" from W11	wall-hung vanity: 2x10 flat, 26"–34" AFF	Jack & Jill double vanity, roll-under (bath face)
B15	W10	1"–3'-0" from south ext. wall	grab bar: 2x8 flat, centre 34" AFF	Jack & Jill toilet, side (bath face)
B16	W16	2"–14'-4" from W15	3/4" ply backing 0"–96" AFF	kitchen tall units, oven, hood (kitchen face)
B17	W18	15'-5"–18'-10" from north ext. wall	rod and shelf: 2x6 flat at 66" and 80" AFF	linen shelves (walk-in face)
B18	W22	24'-10"–30'-2" from west ext. wall	rod and shelf: 2x6 flat at 66" and 80" AFF	walk-in rods

Tag	Wall	Span	Blocking	For
B19	W21	2"–5'-6" from W18	rod and shelf: 2x6 flat at 66" and 80" AFF	walk-in rods (walk-in face)
B20	W13	2"–5'-2" from W7	rod and shelf: 2x6 flat at 66" and 80" AFF	closets 3 and 4, both faces
B21	W9	1"–6'-6" from south ext. wall	rod and shelf: 2x6 flat at 66" and 80" AFF	2nd Suite wardrobe (suite face)
B22	W27	4'-2"–19'-2" from west ext. wall	handrail: 2x6 flat at 34" AFF	stair, open-side handrail
B23	W6	2"–4'-3" from W27	handrail: 2x6 flat at 34" AFF	stair, elevator-hall side
B24	furring	north ext. wall, 10'-0"–16'-0" from east ext. wall	grab bar: 2x8 flat, centre 34" AFF	master shower bench wall (north)
B25	furring	east ext. wall, 2"–3'-4" from W23	grab bar: 2x8 flat, centre 34" AFF	WC toilet, side (east)
B26	furring	north ext. wall, 2"–2'-6" from W20	wall-hung vanity: 2x10 flat, 26"–34" AFF	powder lav, roll-under (north)
B27	furring	south ext. wall, 0"–2'-2" from W10	grab bar: 2x8 flat, centre 34" AFF	Jack & Jill toilet, rear (south)
B28	furring	south ext. wall, 2'-2"–7'-2" from W10	grab bar: 2x8 flat, centre 34" AFF	Jack & Jill tub, long side (south)
B29	furring	east ext. wall, 7'-0"–12'-0" from south ext. wall	TV: 3/4" ply 24" tall, 42"–66" AFF	2nd Suite TV (east)
B30	furring	north ext. wall, 3'-0"–9'-0" from east ext. wall	tub filler: floor-mounted, supply from below	master tub (north)

Bill of materials

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The contractor takes off the final list.

Item	Qty	Notes
Studs 2x4 × 9'	205	16" o.c. plus ends, kings and jacks, +10%
Studs 2x6 × 9'	38	16" o.c. plus ends, kings and jacks, +10%
Studs 2x4 × 9' (staggered)	103	16" o.c. plus ends, kings and jacks, +10%
Studs 2x10 × 9'	15	16" o.c. plus ends, kings and jacks, +10%
Plates 2x4, bottom + double top	619 lf	pressure-treated bottom plate not needed on a wood floor
Plates 2x6, bottom + double top	356 lf	pressure-treated bottom plate not needed on a wood floor
Plates 2x10, bottom + double top	49 lf	pressure-treated bottom plate not needed on a wood floor
Exterior furring 2x4 × 9' @ 24"	136	plus 2x4 plates 464 lf
Flat 2x4 headers with cripples	8	doors D3, D6, D7, D10, D11, D14, D16, D17
Pocket door frame kits, 36" × 96"	8	doors D4, D5, D8, D9, D12, D13, D15, D18; D4, D5 36" (2x4 wall), D18 for a 2x10 wall
Door leaves 8'-0", solid core, lever hardware	15	D4, D5 36"; D3 is a cased opening; D2 by the elevator contractor
2x8 grab-bar blocking	47 lf	centre 34" AFF

Item	Qty	Notes
2x10 vanity blocking	20 lf	26"—34" AFF
3/4" plywood backing (TV, kitchen wall, panel)	6 sheets	4 × 8 sheets
1/2" gypsum board, interior partitions	211 sheets	4 × 8, both faces, +10%; exterior furring board extra: 72 sheets
R-13 / R-19 sound batts	2170 sq ft	bedroom and bath walls, staggered walls R-19
Slip clips for the top plates (if trusses above)	177	one per 2'
Shower floor recess framing	1	2" drop over 6'-6" × 5' plus 12" apron

Quantities are rough-order take-offs from the drawing geometry, for pricing. The contractor takes off the final list.